

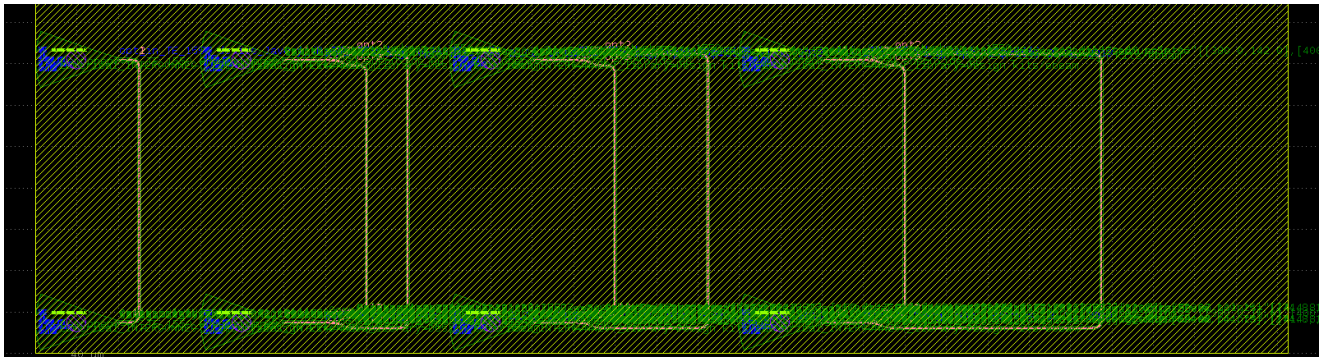
EBeam_Javery – Imbalanced MZIs (TE, 1550 nm)

Designer: Javery · Layout: submissions/EBeam_Javery.oas · Fork: ChenJavery/openEBL-2026-10

Goal: extract the group index n_g of the 500×220 nm strip waveguide from the free spectral range of three MZIs with different ΔL , using $\text{FSR} = \lambda^2 / (n_g \Delta L)$. A GC loopback is included to remove the grating-coupler envelope from the measured spectra.

opt_in label	Structure	ΔL (μm)	Expected FSR @1550 nm
opt_in_TE_1550_device_Javery_LB	GC loopback	—	—
opt_in_TE_1550_device_Javery_MZI1	Y-branch MZI	51	11.2 nm
opt_in_TE_1550_device_Javery_MZI2	Y-branch MZI	101	5.7 nm
opt_in_TE_1550_device_Javery_MZI3	Y-branch MZI	201	2.8 nm

Expected FSR uses $n_g \approx 4.20$ from a MODE simulation of the waveguide. PDK cells: GC_TE_1550_8deg0xide_BB, ebeam_y_1550; waveguide "Strip TE 1550 nm, w=500 nm".



Layout (bottom of the 605×410 μm area). Left to right: loopback, MZI1, MZI2, MZI3.

Verification: SiEPIC layout check 0 errors; GitHub checks passed and the layout is merged (PR #248).